

Natural Language Processing and Large Language Models

NLP & LLMs – IASD App

Course 1

September 28, 2026

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Course Plan

- ▶ Machine learning for natural language processing (NLP)
- ▶ Attention and transformers
- ▶ Transformers and pre-training
- ▶ TP: GPT
- ▶ TP: GPT (Part 2), BERT
- ▶ Evaluation of LLMs

Evaluation:

- 2 homeworks
- 1 graded TP
- 1 project

References and sources:

- Course slides on large language models (LLMs, IASD) by Alexandre Allauzen and Florian Le Bronnec.
- Other references to be detailed.

Evaluation: registration form

Registration : NLP&LLMs course
2026-2027 (IASD App)



Plan du cours

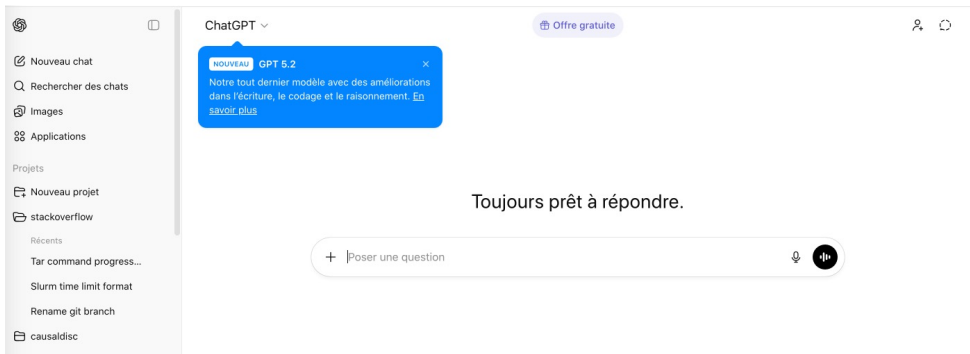
- ▶ 1.1 – LLMs: applications, number of parameters
- ▶ 1.2 – Machine learning for natural language processing (NLP)
- ▶ 1.3 – TP: IMDB dataset, word embeddings

Plan du cours

1.1 – LLMs: applications, number of parameters

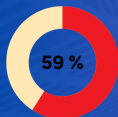
1.2 – Machine learning for natural language processing (NLP)

Large language models: applications

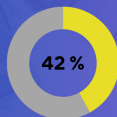


Large language models: applications

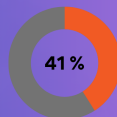
Statistiques d'adoption d'IA dans le secteur bancaire et financier.



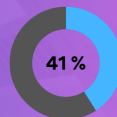
Professionnels
bancaires utilisent l'IA
au quotidien



Responsables de relation
client utilisant l'IA pour
l'automatisation de l'accueil
des clients



Utilisation de l'IA pour le
suivi de la détection des
fraudes

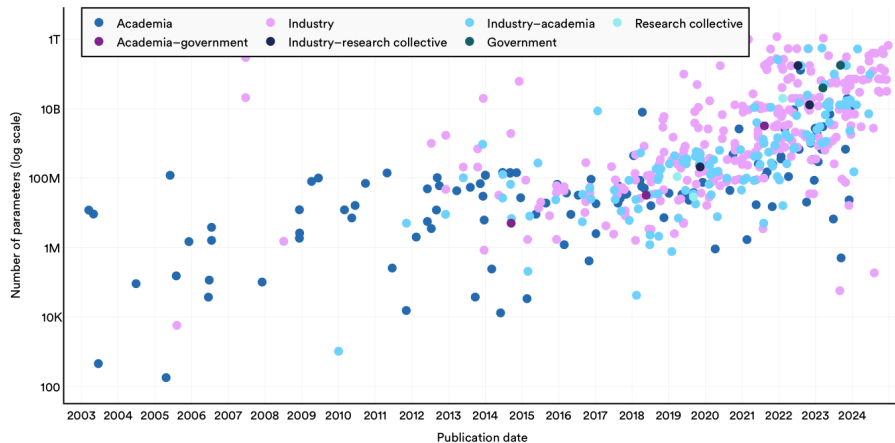


Utilisation de l'IA pour
la gestion des risques
et la conformité

Large language models: number of parameters

Number of parameters of notable AI models by sector, 2003–24

Source: Epoch AI, 2025 | Chart: 2025 AI Index report

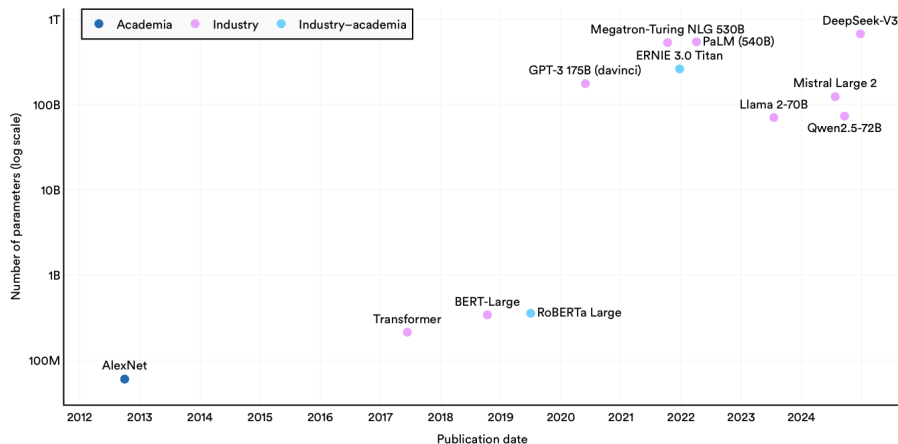


Source: <https://hai.stanford.edu/ai-index/2025-ai-index-report>

Large language models: number of parameters

Number of parameters of select notable AI models by sector, 2012–24

Source: Epoch AI, 2025 | Chart: 2025 AI Index report

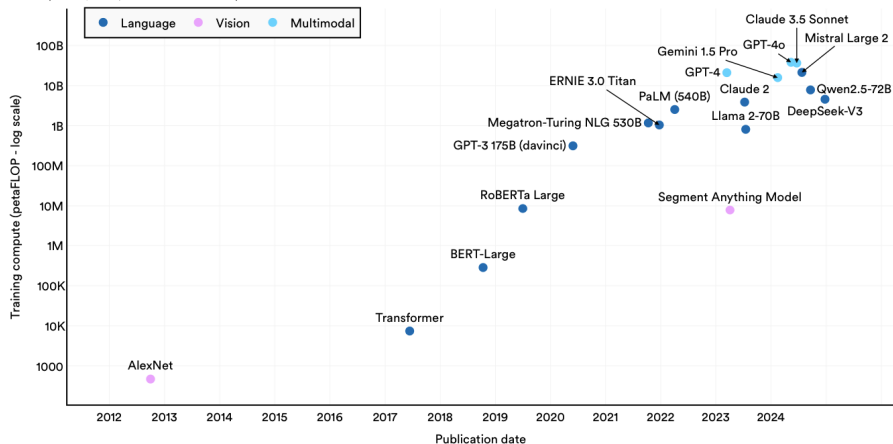


Source: <https://hai.stanford.edu/ai-index/2025-ai-index-report>

Large language models: training compute

Training compute of notable AI models by domain, 2012–24

Source: Epoch AI, 2025 | Chart: 2025 AI Index report



Source: <https://hai.stanford.edu/ai-index/2025-ai-index-report>

Plan du cours

1.1 – LLMs: applications, number of parameters

1.2 – Machine learning for natural language processing (NLP)

Natural language

The goal of natural language (human language) is communication.

► For the speaker:

- Intention: what is the message?
- Generation: transform the message into a sequence.
- Synthesis: output the sequence (speech, written text, signs, ...)

► For the hearer:

- Perception
- Analysis
- Incorporation

Natural language processing: processing of natural language by a computer.

Natural language processing (NLP): challenges

Implicitity: natural language is implicit, which enables conciseness (ellipsis, anaphora, ...), but implicitity entails potential **ambiguities**.

Why does this happen for natural language?

- ▶ Shared world knowledge
- ▶ Cognitive efficiency
- ▶ Social pragmatics

Having a unique linguistic expression for every possible conceptualization would make language overly complex and linguistic expressions unnecessarily long.

(R. Mooney)

Natural language processing (NLP): challenges

Why is natural language ambiguous?

- ▶ Lexical / word-level reasons: synonyms (e.g., bank, plant, . . .)
- ▶ Structural ambiguity (syntax):
e.g., *"I saw the man on the hill with a telescope."*
(From the course of R. Mooney)
- ▶ Referential ambiguity:
e.g., *"Amy told Anna that she was late."*
- ▶ Context dependence:
e.g., *"It's over there."*

This is hard for machines and for AI:

- ▶ No shared knowledge/experience (or lack of shared knowledge for AIs)
- ▶ No physical grounding
- ▶ No social intuition

NLP is a starting point for machines to understand natural language.

Machine learning for NLP

Model NLP tasks as machine learning problems:

- ▶ Text classification

(our focus today)

$$X = (w_1, w_2, \dots, w_L) \rightarrow y(\text{class label})$$

- ▶ Word tagging

$$X = (w_1, w_2, \dots, w_L) \rightarrow (y_1, y_2, \dots, y_L)$$

Text classification

A supervised learning problem:

Training data:

	review	sentiment
0	One of the other reviewers has mentioned that ...	positive
1	A wonderful little production. The...	positive
2	I thought this was a wonderful way to spend ti...	positive
3	Basically there's a family where a little boy ...	negative
4	Petter Mattei's "Love in the Time of Money" is...	positive

Test data:

"My wonderful friend took me to see this movie for our anniversary, it was terrible."

→ negative or positive?

Questions:

- ▶ How to represent the input text?

Unlike images, sounds or other numerical signals, natural language **does not** have straightforward representations in a vector space.

- ▶ How to perform classification?

On Colab:

- ▶ Tokenization
- ▶ Vectorization

Thank You !